ASSOCIATED POINTS OF A FLAT PULLBACK OF A MODULE, AFTER BOURBAKI

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file divisors.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Carrier window: statement and proof both lie inside the window 0-2000 granted to this lane by CONTROL 8.83.

Editorial note. The proof is a genuine reduction: it performs the stalk translation in place and cites the algebra statement only for the translated claim.

Editorial note. The single reference of the counted spans is recorded as a non-load-bearing navigation pointer; the argument that consumes it stays inside the counted proof span.

Editorial note. Preamble: sources/stacks-divisors/retained-preamble.tex (88 lines, verbatim present).

Allowed standard prerequisites (editorial).

- The algebra statement for associated primes of a tensor product over a flat map. Algebra, Lemma algebra-lemma-bourbaki-fibres, used at proof lines 338-341. Is the translation target of the whole proof.

Source reference. Original lemma 05C2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6905–6905. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let $f : X \to S$ be a morphism of schemes. Let $\mathcal{F}$ be a quasi-coherent sheaf on X which is flat over S. Let $\mathcal{G}$ be a quasi-coherent sheaf on S. Then we have*

$$Ass_X(\mathcal{F} \otimes_{\mathcal{O}_X} f^*\mathcal{G}) \supset \bigcup\nolimits_{s \in Ass_S(\mathcal{G})} Ass_{X_s}(\mathcal{F}_s)$$

and equality holds if S is locally Noetherian (for the notation $\mathcal{F}_s$ see above).

Proof. Let $x \in X$ and let $s = f(x) \in S$. Set $B = \mathcal{O}_{X,x}$, $A = \mathcal{O}_{S,s}$, $N = \mathcal{F}_x$, and $M = \mathcal{G}_s$. Note that the stalk of $\mathcal{F} \otimes_{\mathcal{O}_X} f^*\mathcal{G}$ at x is equal to the B-module $M \otimes_A N$. Hence $x \in \mathrm{Ass}_X(\mathcal{F} \otimes_{\mathcal{O}_X} f^*\mathcal{G})$ if and only if $\mathfrak{m}_B$ is in $\mathrm{Ass}_B(M \otimes_A N)$. Similarly $s \in \mathrm{Ass}_S(\mathcal{G})$ and $x \in \mathrm{Ass}_{X_s}(\mathcal{F}_s)$ if and only if $\mathfrak{m}_A \in \mathrm{Ass}_A(M)$ and $\mathfrak{m}_B/\mathfrak{m}_A B \in \mathrm{Ass}_{B \otimes \kappa(\mathfrak{m}_A)}(N \otimes \kappa(\mathfrak{m}_A))$. Thus the lemma follows from Algebra, Lemma 05C2. □

Date: stacks-project:divisors.tex:local-snapshot-20261012; GNU FDL 1.2 or later.